DS4420 Developer Distrust Project - Bayes

Gavin Bond

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Necessary imports:

```
library(readr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(ggplot2)
library(brms)
```

```
## Loading required package: Rcpp

## Loading 'brms' package (version 2.23.0). Useful instructions
## can be found by typing help('brms'). A more detailed introduction
## to the package is available through vignette('brms_overview').

##
## Attaching package: 'brms'

## The following object is masked from 'package:stats':
##
##     ar
```

Initial data import (2 survey years combined). Column types specified due to parsing issue on raw import.

```
df <- readr::read_csv(
  "data/combined_survey.csv",
  col_types = readr::cols(.default = readr::col_character()),
  show_col_types = FALSE
)
```

Ensuring parsing issues were fixed

```r
problems(df)
```

```
## # A tibble: 0 x 5
## # i 5 variables: row <int>, col <int>, expected <chr>, actual <chr>, file <chr>
```

Data inspection 1 (dimensions, top 5 rows, cols)

```r
dim(df)
```

```
## [1] 55118    14
```

```r
head(df)
```

```
## # A tibble: 6 x 14
##   YearsCode WorkExp Age         DevType EdLevel OrgSize AISelect AISent AIThreat
##   <chr>     <chr>   <chr>       <chr>   <chr>   <chr>   <chr>    <chr>  <chr>
## 1 15        <NA>    35-44 year~ Develo~ Master~ <NA>    Yes      Indif~ I'm not~
## 2 12        12.0    25-34 year~ Develo~ Some c~ <NA>    Yes      Very ~ No
## 3 36        <NA>    35-44 year~ Develo~ Master~ <NA>    Yes      Very ~ No
## 4 25        18.0    45-54 year~ Develo~ Bachel~ <NA>    Yes      Very ~ No
## 5 20        17.0    35-44 year~ Develo~ Bachel~ <NA>    Yes      Favor~ I'm not~
## 6 7         6.0     25-34 year~ Cloud ~ Profes~ <NA>    Yes      Very ~ No
## # i 5 more variables: RemoteWork <chr>, AIComplex <chr>, ICorPM <chr>,
## #   AIAcc <chr>, year <chr>
```

```r
names(df)
```

```
##  [1] "YearsCode"  "WorkExp"    "Age"        "DevType"    "EdLevel"
##  [6] "OrgSize"    "AISelect"   "AISent"     "AIThreat"   "RemoteWork"
## [11] "AIComplex"  "ICorPM"     "AIAcc"      "year"
```

Data inspection 2 (summary, checking for NAs)

```r
summary(df)
```

```
##   YearsCode           WorkExp              Age              DevType
##  Length:55118       Length:55118       Length:55118       Length:55118
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##    EdLevel            OrgSize            AISelect            AISent
##  Length:55118       Length:55118       Length:55118       Length:55118
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##    AIThreat           RemoteWork         AIComplex            ICorPM
##  Length:55118       Length:55118       Length:55118       Length:55118
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##     AIAcc               year
##  Length:55118       Length:55118
##  Class :character   Class :character
##  Mode  :character   Mode  :character
```

```r
table(df$AIAcc, useNA = "ifany")
```

```
##
##            Highly distrust                Highly trust
##                       7490                        1366
## Neither trust nor distrust           Somewhat distrust
##                      13582                       13536
##             Somewhat trust
##                      19144
```

```r
table(df$AISelect, useNA = "ifany")
```

```
##
## No, and I don't plan to  No, but I plan to soon                     Yes
##                    3776                    1165                   50154
##                    <NA>
##                      23
```

```r
table(df$year, useNA = "ifany")
```

```
##
##  2024  2025
## 29379 25739
```

```r
colSums(is.na(df))
```

```
##  YearsCode    WorkExp        Age    DevType    EdLevel    OrgSize   AISelect
##       1442      12908          0       1275       1086       6783         23
##     AISent   AIThreat RemoteWork  AIComplex     ICorPM      AIAcc       year
##         39        828       5527        266      16259          0          0
```

From this, we can see there are many missing values; in terms of how it will effect our modeling choices, we are intentionally removing DevType, WorkExp, and ICorPM for our initial fit due to high levels of missing values, high level count, and feature overlap. Will reevaluate this decision after full first model implementation.

Helper function for preprocessing years of experience

```r
# Helper for experience
clean_exp <- function(x) {
  x <- trimws(x)
  out <- dplyr::case_when(
    is.na(x) ~ NA_real_,
    x == "" ~ NA_real_,
    x == "Less than 1 year" ~ 0.5,
    x == "More than 50 years" ~ 51,
    TRUE ~ suppressWarnings(as.numeric(x))
  )
  pmin(out, 51)
}
```

```r
# Helper for other missing factor values
clean_factor <- function(x) {
  x <- trimws(x)
  x[is.na(x) | x == ""] <- "Missing"
  as.factor(x)
}
```

Final inspection for numeric fields

```r
sort(unique(df$YearsCode))[1:10]
```

```
##  [1] "1"     "1.0"   "10"    "10.0"  "100.0" "11"    "11.0"  "12"    "12.0"
## [10] "13"
```

```r
tail(sort(unique(df$YearsCode)), 10)
```

```
##  [1] "7"                  "7.0"                "75.0"
##  [4] "8"                  "8.0"                "82.0"
##  [7] "9"                  "9.0"                "Less than 1 year"
## [10] "More than 50 years"
```

```r
sort(unique(df$WorkExp))[1:10]
```

```
##  [1] "0.0"   "1.0"   "10.0"  "100.0" "11.0"  "12.0"  "13.0"  "14.0"  "15.0"
## [10] "16.0"
```

```r
tail(sort(unique(df$WorkExp)), 10)
```

```
##  [1] "57.0" "58.0" "59.0" "6.0"  "60.0" "65.0" "7.0"  "8.0"  "9.0"  "99.0"
```

Initial model processing: - Restricting model to AI users only - Predicting distrust vs. non-distrust (non-distrust includes "trust" and "neutral") - Using simple feature set for first attempt

```r
df_model <- df %>%
  dplyr::filter(AISelect == "Yes") %>%
  dplyr::transmute(
    y = ifelse(AIAcc %in% c("Somewhat distrust", "Highly distrust"), 1, 0),
    YearsCode = clean_exp(YearsCode),
    Age = clean_factor(Age),
    EdLevel = clean_factor(EdLevel),
    OrgSize = clean_factor(OrgSize),
    AISent = clean_factor(AISent),
    AIThreat = clean_factor(AIThreat),
    RemoteWork = clean_factor(RemoteWork),
    AIComplex = clean_factor(AIComplex),
    year = clean_factor(year)
  )

dim(df_model)
```

```
## [1] 50154    10
```

```r
head(df_model)
```

```
## # A tibble: 6 x 10
##       y YearsCode Age       EdLevel OrgSize AISent AIThreat RemoteWork AIComplex
##   <dbl>     <dbl> <fct>     <fct>   <fct>   <fct>  <fct>    <fct>      <fct>
## 1     0        15 35-44 ye~ Master~ Missing Indif~ I'm not~ Remote     Bad at h~
## 2     0        12 25-34 ye~ Some c~ Missing Very ~ No       Hybrid (s~ Neither ~
## 3     0        36 35-44 ye~ Master~ Missing Very ~ No       In-person  Good, bu~
## 4     0        25 45-54 ye~ Bachel~ Missing Very ~ No       Remote     Good, bu~
## 5     0        20 35-44 ye~ Bachel~ Missing Favor~ I'm not~ Remote     Good, bu~
## 6     0         7 25-34 ye~ Profes~ Missing Very ~ No       Remote     Bad at h~
## # i 1 more variable: year <fct>
```

```r
table(df_model$y)
```

```
##
##     0     1
## 33064 17090
```

Final preprocessing for model

```r
summary(df_model$YearsCode)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##    0.50    8.00   12.00   15.19   20.00   51.00    1420
```

```r
sum(is.na(df_model$YearsCode))
```

```
## [1] 1420
```

```r
df_model <- df_model %>%
  dplyr::filter(!is.na(YearsCode))

df_model$YearsCode_z <- as.numeric(scale(df_model$YearsCode))

summary(df_model$YearsCode_z)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -1.4524 -0.7107 -0.3152  0.0000  0.4760  3.5415
```

```r
dim(df_model)
```

```
## [1] 48734    11
```

```r
table(df_model$y)
```

```
##
##     0     1
## 31931 16803
```

Final modeling df

```r
df_model <- df_model %>%
  dplyr::select(
    y, YearsCode_z, Age, EdLevel, OrgSize,
    AISent, AIThreat, RemoteWork, AIComplex, year
  )

str(df_model)
```

```
## tibble [48,734 x 10] (S3: tbl_df/tbl/data.frame)
##  $ y          : num [1:48734] 0 0 0 0 0 0 0 1 1 0 ...
##  $ YearsCode_z: num [1:48734] -0.0185 -0.3152 2.0582 0.9704 0.476 ...
##  $ Age        : Factor w/ 8 levels "18-24 years old",..: 3 2 3 4 3 2 2 7 2 2 ...
##  $ EdLevel    : Factor w/ 10 levels "Associate degree (A.A., A.S., etc.)",..: 3 9 3 2 2 7 2 10 1 7 .
##  $ OrgSize    : Factor w/ 10 levels "1,000 to 4,999 employees",..: 10 10 10 10 10 10 10 10 10 10 ...
##  $ AISent     : Factor w/ 7 levels "Favorable","Indifferent",..: 2 6 6 6 1 6 6 2 2 1 ...
##  $ AIThreat   : Factor w/ 4 levels "I'm not sure",..: 1 3 3 3 1 3 3 3 3 3 ...
##  $ RemoteWork : Factor w/ 4 levels "Hybrid (some remote, some in-person)",..: 4 1 2 4 4 4 4 3 2 4 ..
##  $ AIComplex  : Factor w/ 7 levels "Bad at handling complex tasks",..: 1 5 2 2 2 1 2 5 5 1 ...
##  $ year       : Factor w/ 2 levels "2024","2025": 1 1 1 1 1 1 1 1 1 1 ...
```

Initial bayes log-regression formula

```r
init_bayes_formula <- bf(
  y ~ YearsCode_z + AISent + AIThreat + AIComplex + year
)

# Default prior
init_def_prior <- default_prior(
  init_bayes_formula,
  data = df_model,
  family = bernoulli("logit")
)

init_def_prior
```

```
##                  prior     class
##                 (flat)         b
##                 (flat)         b
##                 (flat)         b
##                 (flat)         b
##                 (flat)         b
##                 (flat)         b
##                 (flat)         b
##                 (flat)         b
##                 (flat)         b
##                 (flat)         b
##                 (flat)         b
##                 (flat)         b
##                 (flat)         b
##                 (flat)         b
##                 (flat)         b
##                 (flat)         b
##                 (flat)         b
```

```
##                (flat)         b
##                (flat)         b
##  student_t(3, 0, 2.5) Intercept
##                                                 coef group resp dpar nlpar lb ub
##
##     AIComplexGoodbutnotgreatathandlingcomplextasks
##  AIComplexIdontuseAItoolsforcomplextasksDIdontknow
##                                   AIComplexMissing
##    AIComplexNeithergoodorbadathandlingcomplextasks
##            AIComplexVerypoorathandlingcomplextasks
##            AIComplexVerywellathandlingcomplextasks
##                                  AISentIndifferent
##                                      AISentMissing
##                                  AISentUnfavorable
##                                       AISentUnsure
##                                AISentVeryfavorable
##                              AISentVeryunfavorable
##                                    AIThreatMissing
##                                         AIThreatNo
##                                        AIThreatYes
##                                           year2025
##                                        YearsCode_z
##
##  tag       source
##           default
##      (vectorized)
##      (vectorized)
##      (vectorized)
##      (vectorized)
##      (vectorized)
##      (vectorized)
##      (vectorized)
##      (vectorized)
##      (vectorized)
##      (vectorized)
##      (vectorized)
##      (vectorized)
##      (vectorized)
##      (vectorized)
##      (vectorized)
##      (vectorized)
##      (vectorized)
##      (vectorized)
##           default
```

```r
# Manual prior
init_manual_prior <- c(
  brms::set_prior("student_t(4, 0, 10)", class = "Intercept"),
  brms::set_prior("normal(0, 2)", class = "b")
)
```

Fitting initial bayes log-regression model

```r
init_brm_logit <- brms::brm(
  formula = init_bayes_formula,
  family = bernoulli("logit"),
  data = df_model,
  chains = 4,
  iter = 1000,
  warmup = 180,
  prior = init_manual_prior,
  seed = 4420
)
```

```
## Compiling Stan program...

## Trying to compile a simple C file

## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 17.0.0 (clang-1700.6.3.2)'
## using SDK: 'MacOSX26.2.sdk'
## clang -arch arm64 -std=gnu2x -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG   -I"/Li
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/StanHeade
## In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/RcppEigen
## In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/RcppEigen
## /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/RcppEigen/include/Eigen/src/Cor
##   679 | #include <cmath>
##       |          ^~~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1

## Start sampling
```

Checking initial model results & diagnostics

```r
summary(init_brm_logit)
```

```
##  Family: bernoulli
##   Links: mu = logit
## Formula: y ~ YearsCode_z + AISent + AIThreat + AIComplex + year
##    Data: df_model (Number of observations: 48734)
##   Draws: 4 chains, each with iter = 1000; warmup = 180; thin = 1;
##          total post-warmup draws = 3280
##
## Regression Coefficients:
##                                                  Estimate Est.Error l-95% CI
## Intercept                                           -0.55      0.03    -0.61
## YearsCode_z                                          0.18      0.01     0.16
## AISentIndifferent                                    0.80      0.03     0.74
## AISentMissing                                       -0.20      0.48    -1.19
## AISentUnfavorable                                    2.09      0.06     1.97
## AISentUnsure                                         0.92      0.09     0.74
## AISentVeryfavorable                                 -0.69      0.03    -0.75
```

```
## AISentVeryunfavorable                                  2.37      0.14     2.10
## AIThreatMissing                                       -0.16      0.18    -0.52
## AIThreatNo                                             0.23      0.03     0.17
## AIThreatYes                                            0.08      0.04    -0.01
## AIComplexGoodbutnotgreatathandlingcomplextasks        -1.72      0.03    -1.79
## AIComplexIdontuseAItoolsforcomplextasksDIdontknow     -0.02      0.07    -0.16
## AIComplexMissing                                      -0.85      0.18    -1.20
## AIComplexNeithergoodorbadathandlingcomplextasks       -1.09      0.03    -1.15
## AIComplexVerypoorathandlingcomplextasks                1.16      0.03     1.09
## AIComplexVerywellathandlingcomplextasks               -2.44      0.12    -2.69
## year2025                                               0.16      0.03     0.11
##                                                   u-95% CI Rhat Bulk_ESS
## Intercept                                            -0.48 1.00     3136
## YearsCode_z                                           0.21 1.00     6013
## AISentIndifferent                                     0.86 1.00     4401
## AISentMissing                                         0.70 1.00     4462
## AISentUnfavorable                                     2.21 1.00     4159
## AISentUnsure                                          1.10 1.00     3550
## AISentVeryfavorable                                  -0.63 1.00     3785
## AISentVeryunfavorable                                 2.64 1.00     4362
## AIThreatMissing                                       0.20 1.00     3089
## AIThreatNo                                            0.29 1.00     3000
## AIThreatYes                                           0.16 1.00     2798
## AIComplexGoodbutnotgreatathandlingcomplextasks       -1.66 1.00     3133
## AIComplexIdontuseAItoolsforcomplextasksDIdontknow     0.13 1.00     4705
## AIComplexMissing                                     -0.52 1.00     5240
## AIComplexNeithergoodorbadathandlingcomplextasks      -1.02 1.00     3241
## AIComplexVerypoorathandlingcomplextasks               1.23 1.00     3255
## AIComplexVerywellathandlingcomplextasks              -2.21 1.00     3761
## year2025                                              0.21 1.00     4540
##                                                   Tail_ESS
## Intercept                                             2468
## YearsCode_z                                           2410
## AISentIndifferent                                     2736
## AISentMissing                                         2132
## AISentUnfavorable                                     2622
## AISentUnsure                                          2298
## AISentVeryfavorable                                   2435
## AISentVeryunfavorable                                 2339
## AIThreatMissing                                       2250
## AIThreatNo                                            2679
## AIThreatYes                                           2788
## AIComplexGoodbutnotgreatathandlingcomplextasks        2650
## AIComplexIdontuseAItoolsforcomplextasksDIdontknow     2606
## AIComplexMissing                                      2420
## AIComplexNeithergoodorbadathandlingcomplextasks       2479
## AIComplexVerypoorathandlingcomplextasks               2823
## AIComplexVerywellathandlingcomplextasks               2454
## year2025                                              2207
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

Checking initial model diagnostics plot

```
#plot(init_brm_logit)
```

Posterior predicted probabilities for initial model

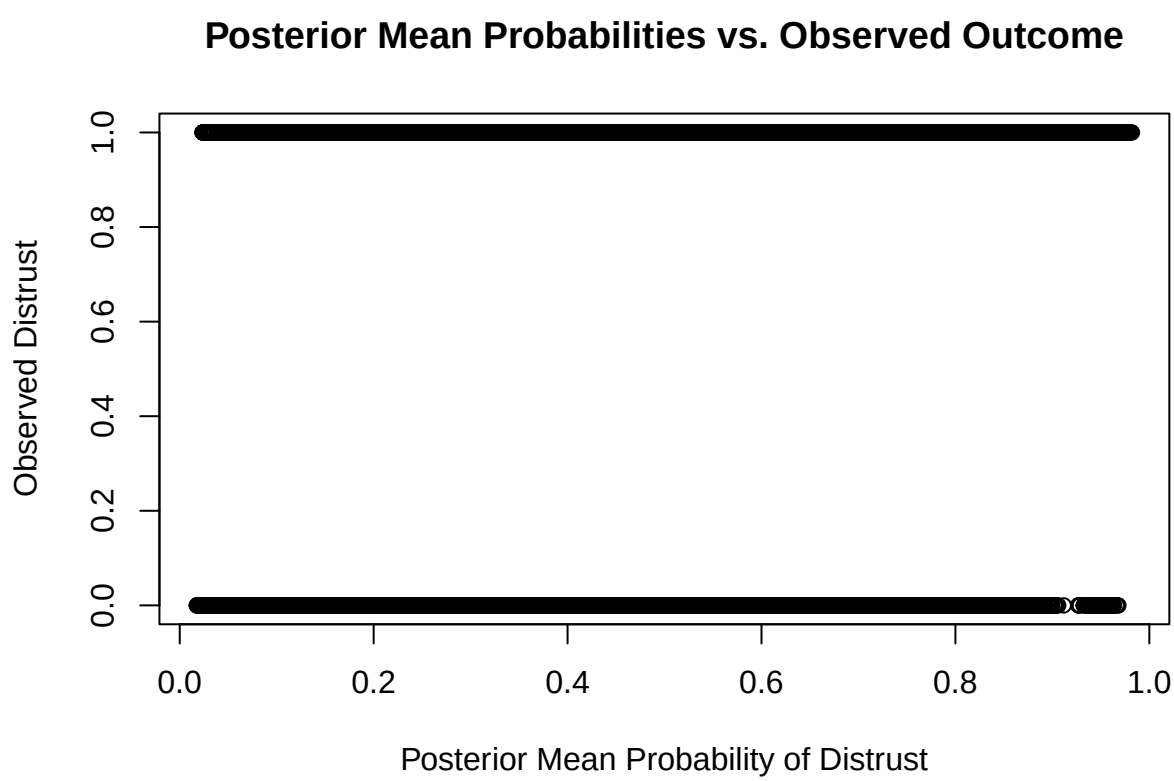
```
# Computing predictions
init_pred_probs <- fitted(init_brm_logit)[, "Estimate"]

# Viewing prediction sample and summary
head(init_pred_probs)
```

```
## [1] 0.56098657 0.10356511 0.08623503 0.07178348 0.10120917 0.23816843
```

```
summary(init_pred_probs)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01765 0.10540 0.25496 0.34481 0.53966 0.98208
```

```
# Predictions vs. reality plot
plot(init_pred_probs, df_model$y,
     main = "Posterior Mean Probabilities vs. Observed Outcome",
     xlab = "Posterior Mean Probability of Distrust",
     ylab = "Observed Distrust")
```



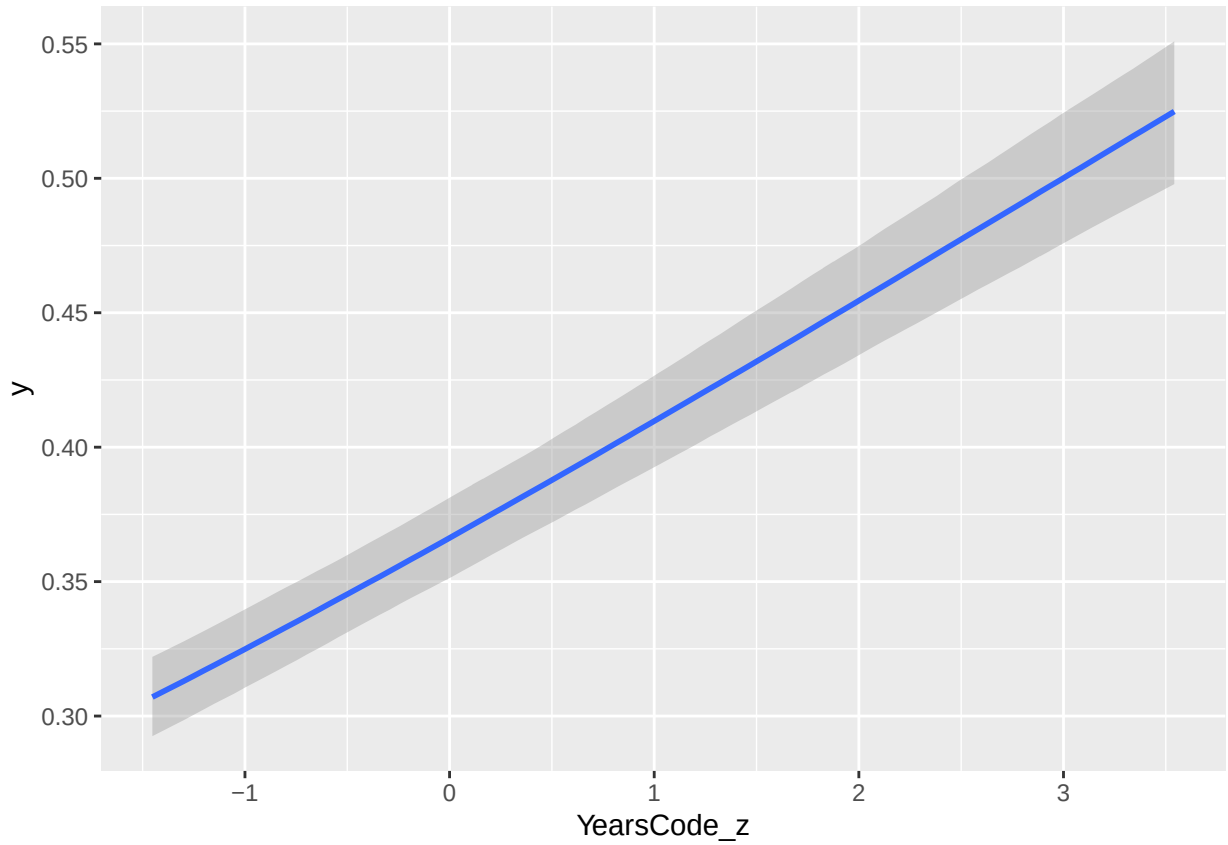
Improved summary output and plots

```r
fixef(init_brm_logit)
```

```
##                                                   Estimate  Est.Error
## Intercept                                       -0.54876108 0.03262393
## YearsCode_z                                      0.18290304 0.01158420
## AISentIndifferent                                0.79739982 0.03012999
## AISentMissing                                   -0.19850761 0.48311016
## AISentUnfavorable                                2.09287417 0.06184207
## AISentUnsure                                     0.91555366 0.09304467
## AISentVeryfavorable                             -0.69258572 0.03160468
## AISentVeryunfavorable                            2.36658502 0.13944888
## AIThreatMissing                                 -0.15741140 0.18314434
## AIThreatNo                                       0.22635988 0.03008803
## AIThreatYes                                      0.07555146 0.04295854
## AIComplexGoodbutnotgreatathandlingcomplextasks  -1.72279574 0.03256308
## AIComplexIdontuseAItoolsforcomplextasksDIdontknow -0.01972991 0.07279061
## AIComplexMissing                                -0.85368930 0.17603370
## AIComplexNeithergoodorbadathandlingcomplextasks -1.08618042 0.03166614
## AIComplexVerypoorathandlingcomplextasks          1.16276895 0.03492406
## AIComplexVerywellathandlingcomplextasks         -2.44095879 0.12311081
## year2025                                         0.15927830 0.02530937
##                                                        Q2.5      Q97.5
## Intercept                                       -0.61312538 -0.4845926
## YearsCode_z                                      0.16082589  0.2057224
## AISentIndifferent                                0.73839222  0.8561258
## AISentMissing                                   -1.19328480  0.7046000
## AISentUnfavorable                                1.97084399  2.2127003
## AISentUnsure                                     0.74108924  1.0990455
## AISentVeryfavorable                             -0.75475779 -0.6315131
## AISentVeryunfavorable                            2.10365663  2.6446512
## AIThreatMissing                                 -0.51769748  0.1964865
## AIThreatNo                                       0.16781396  0.2864923
## AIThreatYes                                     -0.01103423  0.1585437
## AIComplexGoodbutnotgreatathandlingcomplextasks  -1.78775594 -1.6574988
## AIComplexIdontuseAItoolsforcomplextasksDIdontknow -0.15744660  0.1264893
## AIComplexMissing                                -1.19904110 -0.5190045
## AIComplexNeithergoodorbadathandlingcomplextasks -1.15040163 -1.0244263
## AIComplexVerypoorathandlingcomplextasks          1.09477564  1.2299552
## AIComplexVerywellathandlingcomplextasks         -2.68594284 -2.2085613
## year2025                                         0.10816263  0.2091552
```

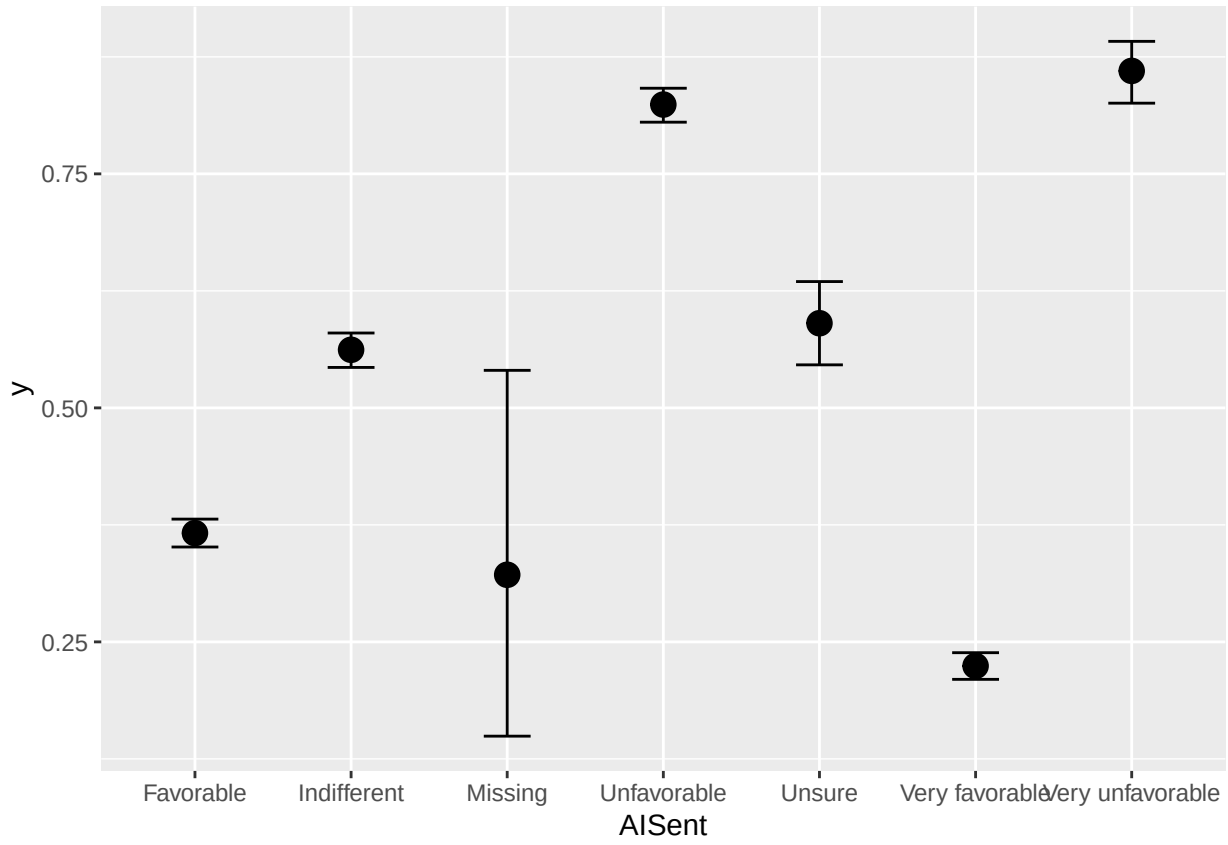
```r
plot(conditional_effects(init_brm_logit, "YearsCode_z"))
```

```
## Ignoring unknown labels:
## * fill : "NA"
## * colour : "NA"
## Ignoring unknown labels:
## * fill : "NA"
## * colour : "NA"
```



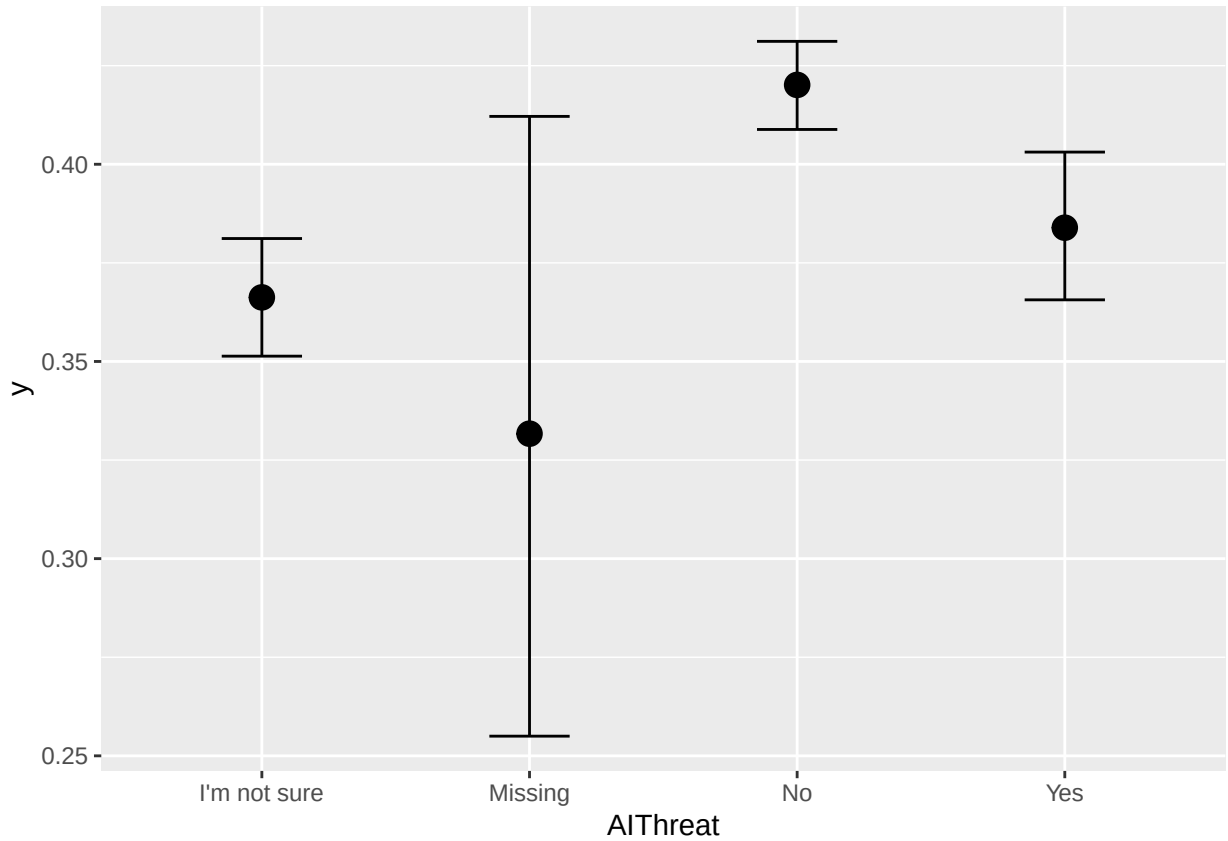

```r
plot(conditional_effects(init_brm_logit, "AISent"))
```

```
## Ignoring unknown labels:
## * fill : "NA"
## * colour : "NA"
## Ignoring unknown labels:
## * fill : "NA"
## * colour : "NA"
```



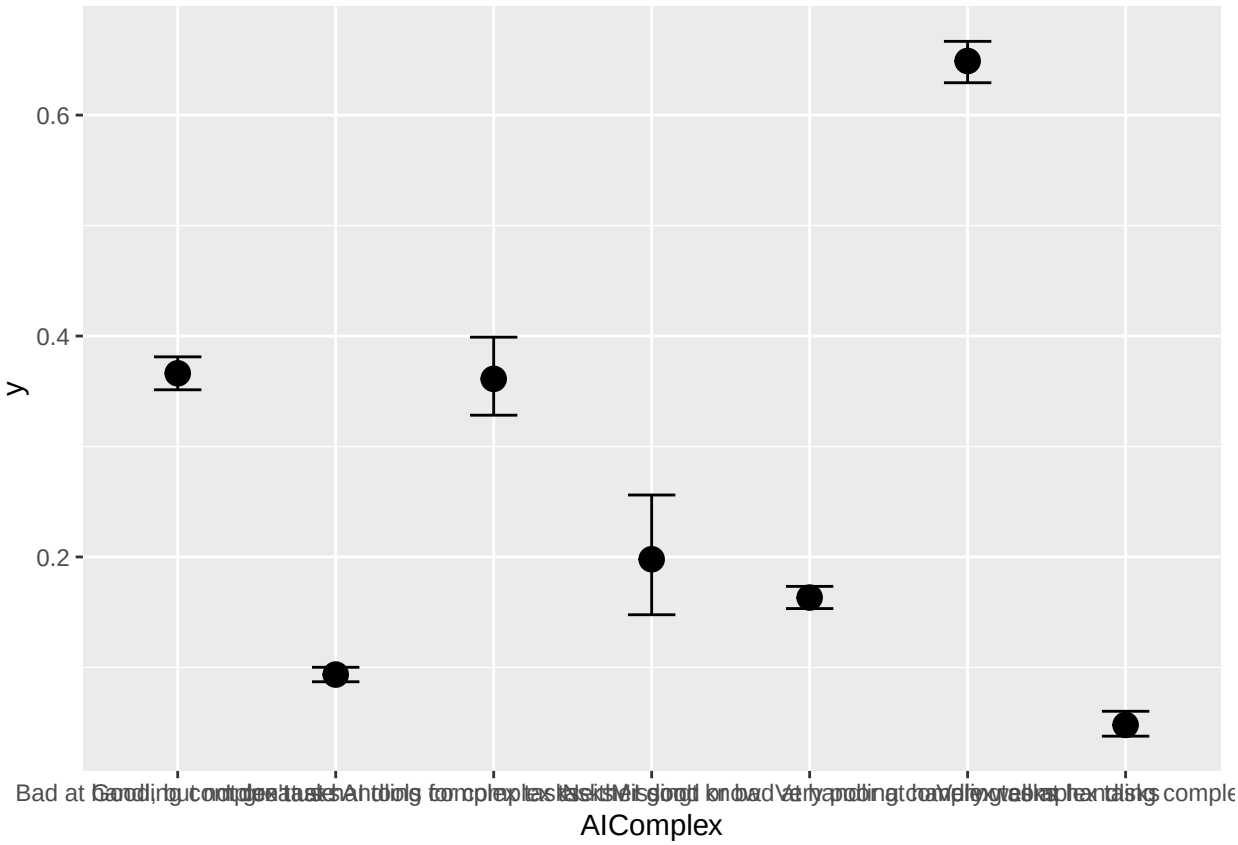

```r
plot(conditional_effects(init_brm_logit, "AIThreat"))
```

```
## Ignoring unknown labels:
## * fill : "NA"
## * colour : "NA"
## Ignoring unknown labels:
## * fill : "NA"
## * colour : "NA"
```



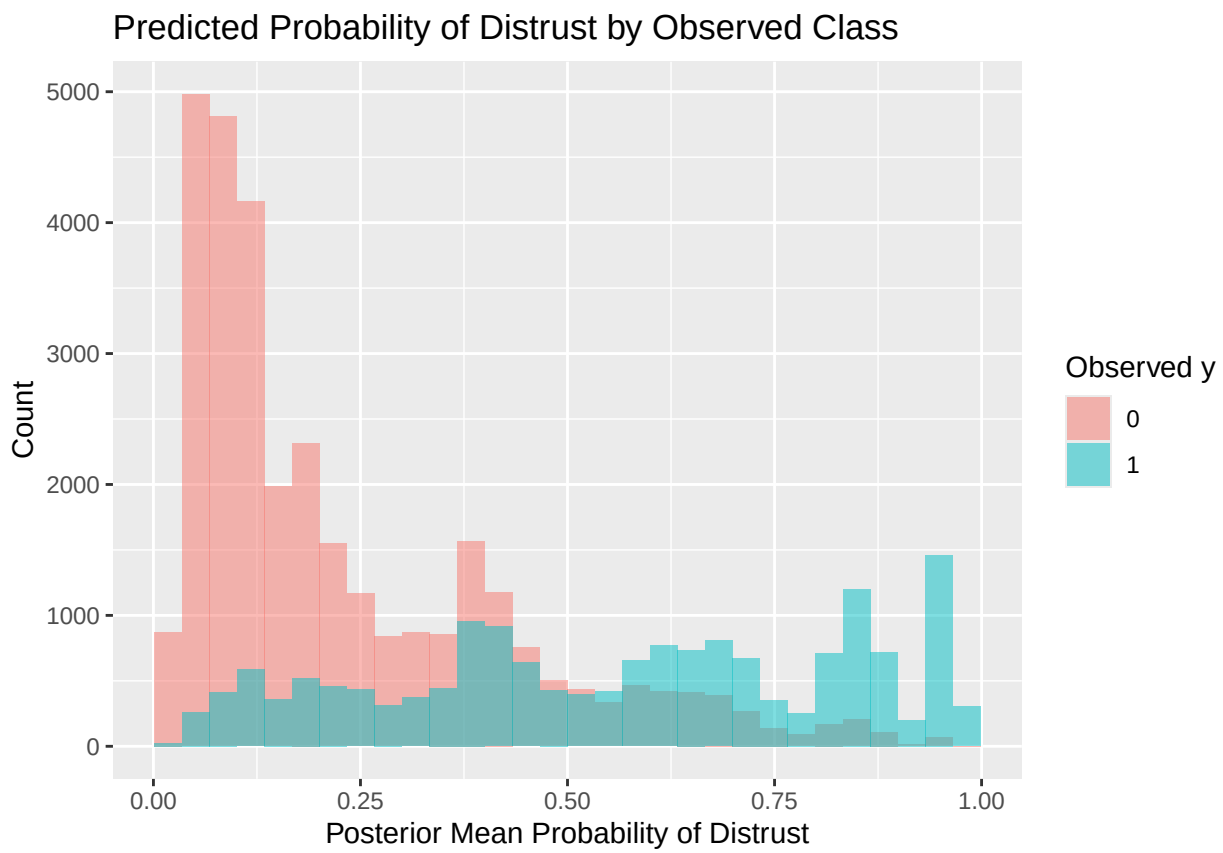

```r
plot(conditional_effects(init_brm_logit, "AIComplex"))
```

```
## Ignoring unknown labels:
## * fill : "NA"
## * colour : "NA"
## Ignoring unknown labels:
## * fill : "NA"
## * colour : "NA"
```



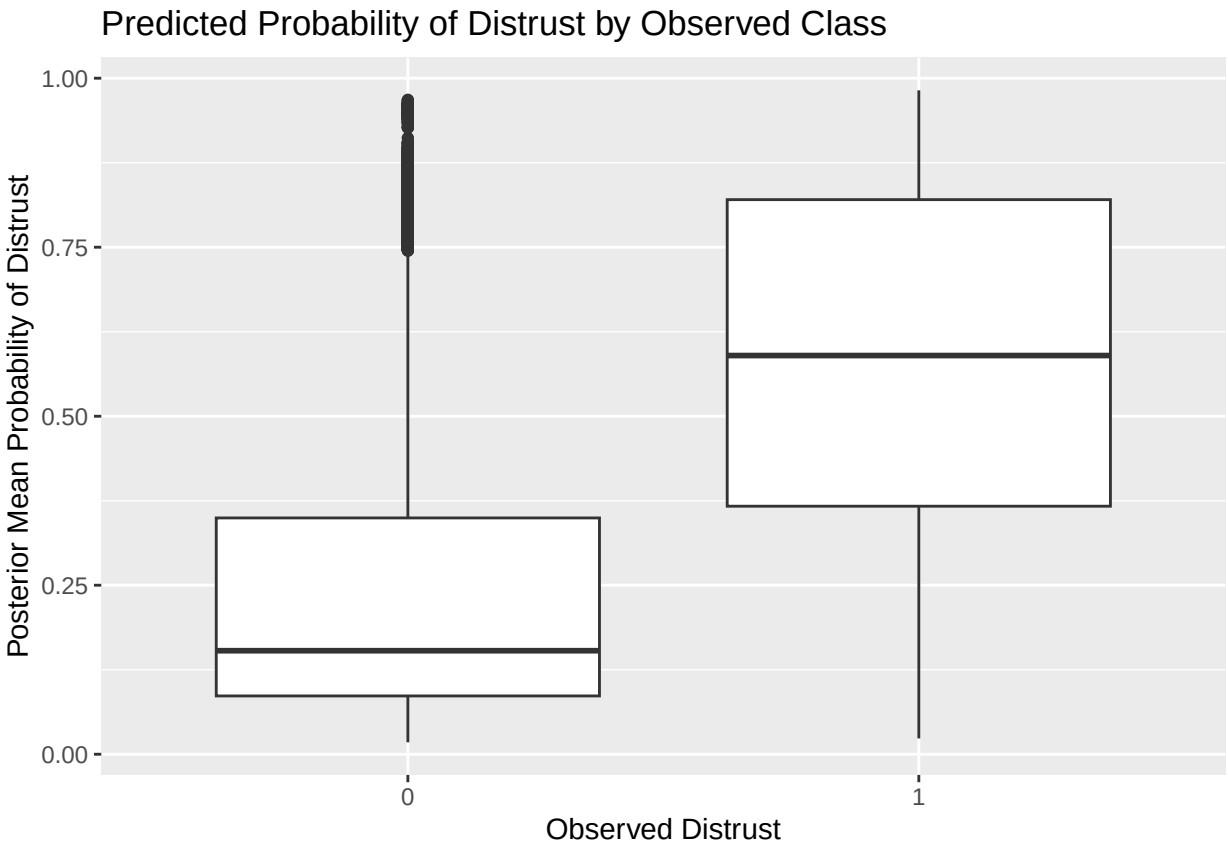
Predicted probabilities by observed class

```
prob_df <- data.frame(
  prob = init_pred_probs,
  y = as.factor(df_model$y)
)

# New plot, posterior mean prob. vs actual count
ggplot(prob_df, aes(x = prob, fill = y)) +
  geom_histogram(position = "identity", alpha = 0.5, bins = 30) +
  labs(
    title = "Predicted Probability of Distrust by Observed Class",
    x = "Posterior Mean Probability of Distrust",
    y = "Count",
    fill = "Observed y"
  )
```



Boxplot of predicted probabilities by class

```
ggplot(prob_df, aes(x = y, y = prob)) +
  geom_boxplot() +
  labs(
    title = "Predicted Probability of Distrust by Observed Class",
    x = "Observed Distrust",
    y = "Posterior Mean Probability of Distrust"
  )
```


Predicted Probability of Distrust by Observed Class
Posterior Mean Probability of Distrust
1.00
0.75
0.50
0.25
0.00
0
1
Observed Distrust